Extractive Question Answering on Course Material using a Pretrained Transformer Encoder

AI Mini-Capstone | Artificial Intelligence (25CS301) | Topic 3.4 LLMs and Transformer Models

Team: Team 4: Capstone AI

PROBLEM STATEMENT

Finding a short fact (e.g. *"Which property makes A* optimal?"*) in long lecture notes is slow. Keyword search returns whole pages or sentences, not the answer. **Goal:** given a question, return the exact answer span from the course material, with its source and a confidence score.

OBJECTIVES

1. Build an **extractive QA** system over AI course notes.
2. Use a **pretrained transformer encoder** (DistilBERT) to predict answer start/end positions.
3. Compare it with a TF-IDF sentence-retrieval baseline using EM, F1 and Hit@k.

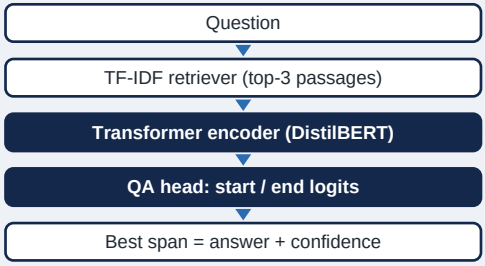
AI TECHNIQUE / ALGORITHM

Reader: transformer encoder (self-attention, 6 layers, 12 heads) fine-tuned on SQuAD. Input `[CLS] question [SEP] passage [SEP]`. A linear QA head outputs a **start logit and an end logit per token**; softmax gives probabilities; the answer is the span maximising P(start) x P(end) inside the passage. **Retriever:** TF-IDF + cosine similarity selects the top-3 passages; the most confident span wins.

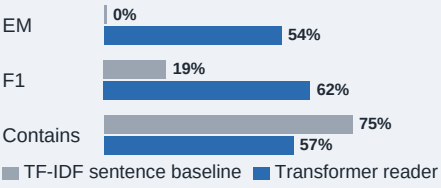
IMPLEMENTATION

Python, PyTorch, Hugging Face Transformers, scikit-learn. Corpus: **21 passages** of AI course notes (search, knowledge-based agents, planning, LLMs); **28 hand-written test questions** with gold answers. Command-line demo prints the answer, confidence and source passage.

METHODOLOGY / WORKFLOW



KEY RESULTS



Retriever Hit@3: **100%** | Latency: **264 ms**/question | 28 test questions

ANALYSIS

Baseline returns whole sentences, so EM is low (0%) even when the right sentence is found (75% contain the answer). The transformer reader returns a short span (EM 54%, F1 62%). Remaining errors are listed in the report.

REAL-WORLD APPLICATIONS & LIMITATIONS

Uses: study assistants, FAQ and helpdesk bots, document search, and the reader stage of RAG pipelines. **Limits:** answer must exist as a span in the text; it cannot combine facts or explain; a SQuAD-trained model can be confidently wrong; small test set.

CONCLUSION / FUTURE SCOPE

Result: EM **54%**, F1 **62%** vs baseline EM 0%, F1 19%. **Next:** dense retrieval (sentence embeddings), SQuAD 2.0 no-answer handling, fine-tuning on course Q&A, generative RAG with citations.